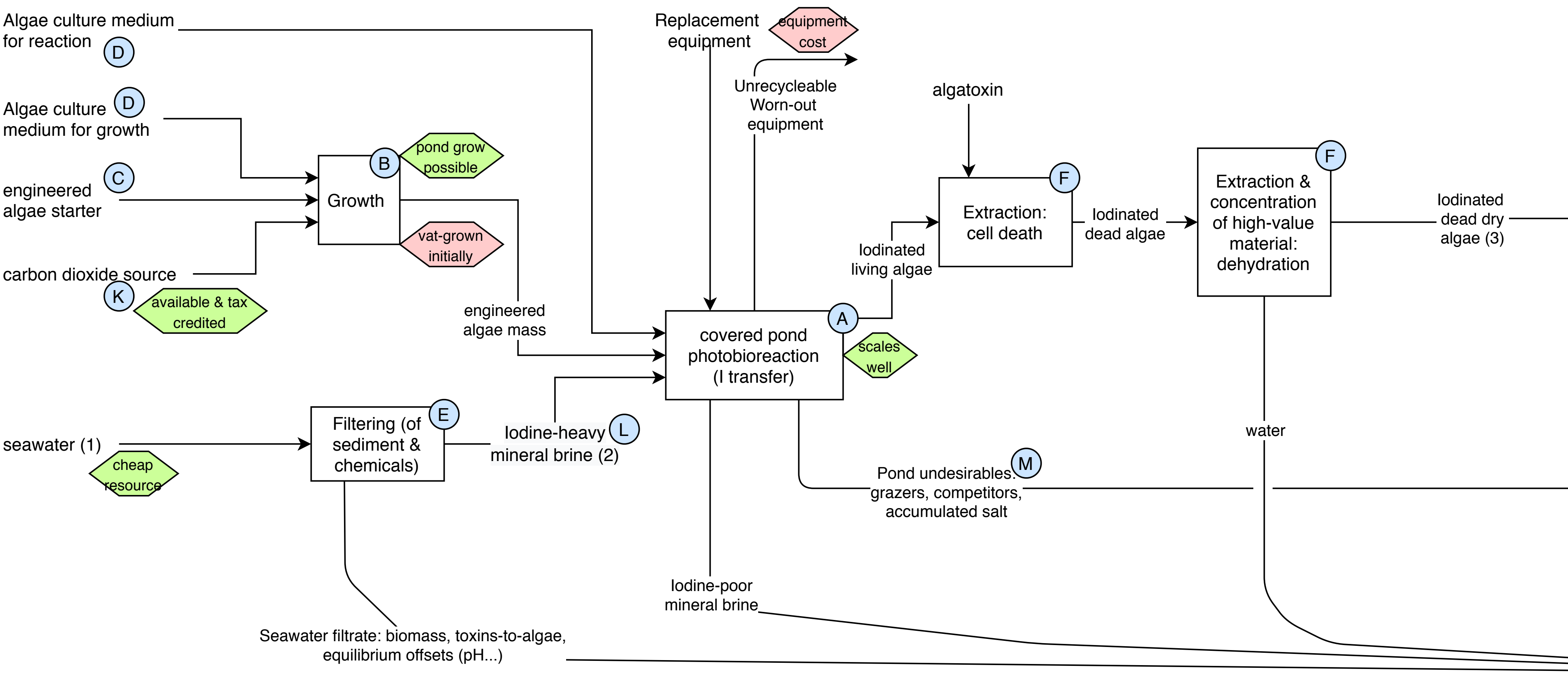
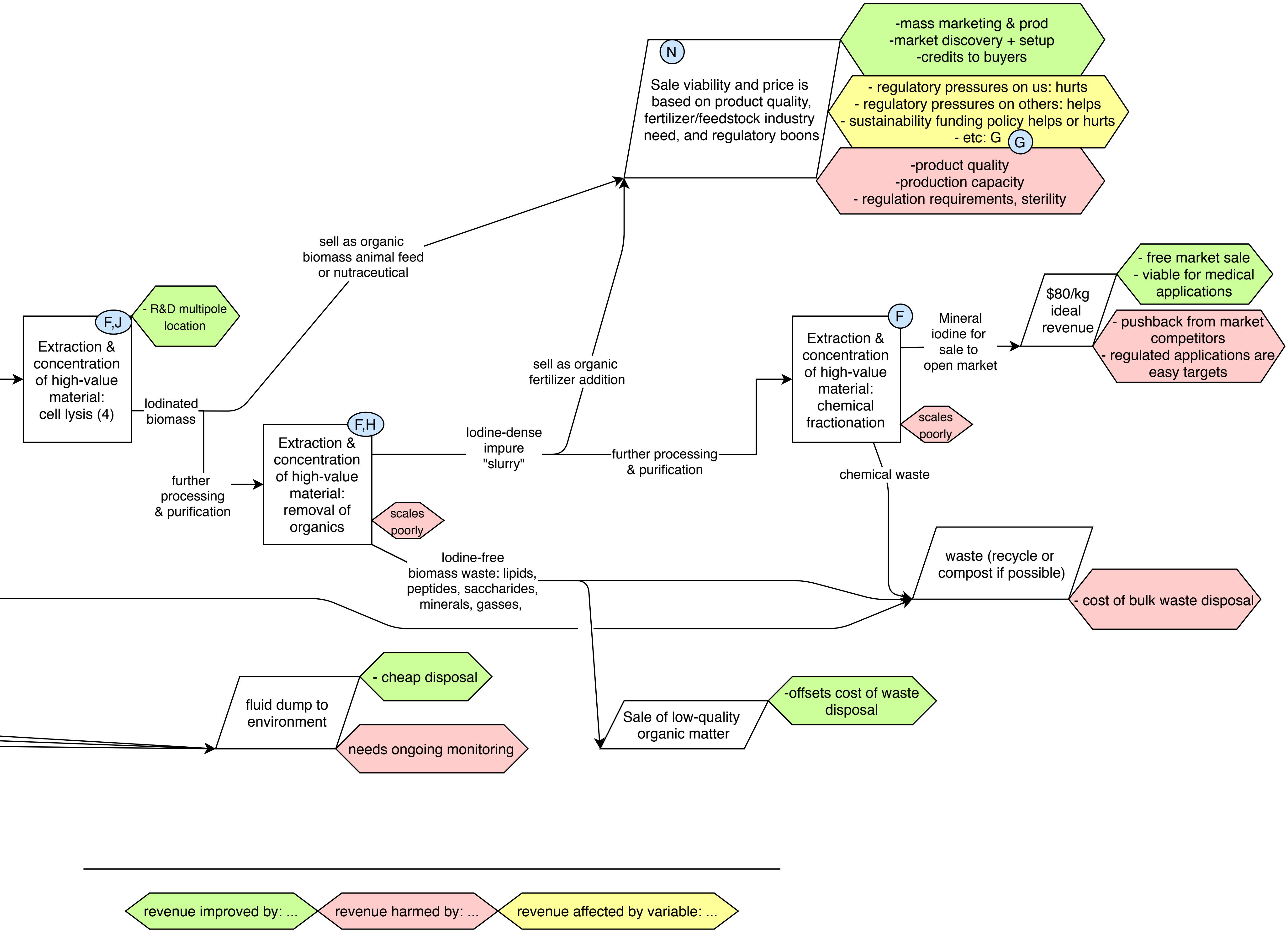
Block flow diagram of iodine harvesting, May 2026, Morgan
thanks to youtube's LearnChemE video on "Characteristics of BFDs"

Industrial loop



Industrial loop notes.

During R&D, (2) is cheaper. At scale, (1) plus filtering is cheaper. (3): cell death and dehydration might reverse in process's order, as well as with (4), although if R&D develops a method for extracellular accumulation of iodine, (4) may not be necessary. Most of the extraction steps can be outsourced at the cost of transport but the relief of internally setting up refining. Main financial factors and viability depends on colorful hexagons (to right). At stable scale, recurring costs are media, seawater filtering and power for pumping, equipment maintenance, power & speciality inputs for reactors, power & bulk inputs for downstream extraction, transport and logistics of final product, quality control, staffing, cumulative impacts on the environmental vicinity, and possible policy volatility. If those cost less than the final revenue and the setup and "prorated" R&D overhead, then this is a viable business. Property protection can be maintained via R&D at points C, E, securing K and locking them in, the algatoxin, temporarily with sale contracts, and detailed points in F that aren't obvious.


Extraction & concentration of high-value material: cell lysis (4)
F,J
- R&D multipole location
Iodinated biomass
further processing & purification
Extraction & concentration of high-value material: removal of organics
F,H
scales poorly
sell as organic biomass animal feed or nutraceutical
N
Sale viability and price is based on product quality, fertilizer/feedstock industry need, and regulatory boons
-mass marketing & prod
-market discovery + setup
-credits to buyers
- regulatory pressures on us: hurts
- regulatory pressures on others: helps
- sustainability funding policy helps or hurts
- etc: G
G
-product quality
-production capacity
- regulation requirements, sterility
sell as organic fertilizer addition
Iodine-dense impure "slurry"
further processing & purification
Extraction & concentration of high-value material: chemical fractionation
F
scales poorly
Mineral iodine for sale to open market
$80/kg ideal revenue
- free market sale
- viable for medical applications
- pushback from market competitors
- regulated applications are easy targets
chemical waste
Iodine-free biomass waste: lipids, peptides, saccharides, minerals, gasses,
waste (recycle or compost if possible)
- cost of bulk waste disposal
fluid dump to environment
- cheap disposal
needs ongoing monitoring
Sale of low-quality organic matter
-offsets cost of waste disposal
revenue improved by: ...
revenue harmed by: ...
revenue affected by variable: ...

Research Development loops

A

A: covered pond R&D. This will take mostly experimental setups; previous work exists but is limited.

It costs rental/ownership of dry land near seawater, regulatory permits and maintenance, design, and construction costs.

Actual materials are low-cost. However, costs of iteratively developing functioning ponds could run very expensive without results to justify funding rounds.

B

B: scale-up bioreactor. Most of the design can be contracted out, but costs of acquiring and running the reactor are substantial. Also needs reactor design expertise. If strain or species changes, then reactor will need to change, so flexible-settings reactors are ideal during development. Scale-up is always a concern.

Alternatively, the algae could in theory be grown in a separate secondary open pond system, for mere biomass, if it's economical.

C

C: engineering of select microbe. This is somewhat expensive and requires lab access, but can be done relatively independently or in school. Viability-discovery and optimization are different tasks, but changes here affect many process parts downstream. So a lot of focus needs to go into making good design choices at this stage.

D

D: Growth medium. This is likely a proprietary purchase that will be upcharged. Medium for growth needs technical precision, and medium for reaction also must be equilibriated. Media for well-studied species may be cheaper to get.

H

H: extraction of cell organics. There might be tons of biofuel people ready to contribute expertise to this part.

J

J: R&D multipole location. Multiple products from one upstream process. Other engineered products of value can be grown using the same upstream up to this point, meaning R&D innovations costs only one additional separation or extraction.

K

K: CO_2 input. This is an important part of the process, as it can be both piped in from exhaust, from industrial CO_2 tanks, and potentially provided directly from nature. It underpins a whole category of funding, carbon capture. It would be wonderful to run this whole process on "waste" products!

See Algal Biorefining, ~page 10 for cliffnotes on what kind of exhaust scrubbing would be needed prior to feeding a CO_2-heavy exhaust to an algae.

L

L: Iodine-heavy mineral water or brine can be made in lab cheaply. Scaling this up to include upstream seawater is definitely an expensive part of the design process, and requires a marine ecologist with expertise about local conditions to get right.

M

M: pond undesirables and environmental contamination can be mitigated with filtration (E) and by covering the ponds with transparent but an air- and contaminant-impervious layer roof. They don't have to be perfectly imporous, of course, but that is a process efficiency for later in development.

E

E: Filtering (of sediment & chemicals) may get more or less expensive across scale and development. Increasing the amount of filter meshes and catalysts is linearly expensive, but engineering microbe to withstand more de-novo seawater might be both possible or even beneficial to the organism. Being upstream, tweaks here have lasting effects, and improvements here save costs down the line, especially regarding desalination to preserve equipment lifespan.

Also, this could be presented as a bioremediation process for funding & permitting.

F

F: Downstream extraction and purification. This is definitely expensive, but benefits from being a not-novel procedure that's been optimized. Much of the work is time-worn (bio)chemical separation, which is expensive in energy and at scale of equipment, but can likely be well-estimated given decades of prior implementations. Need a to involve a pricey process chemical engineer in-house or in contract to make this possible. Finally, this needs to be more efficient than kelp harvesting, probably via growth rate.

G

G: "advantage" sales. If product can be sold before total purification, then (a) costs of intensive further processing can be reduced and (b) focus can be placed on everything upstream of here. Basically, a product midway through the pipeline buys resources to not worry as much about what's further downstream. Plus, these products are more amenable to getting grant funding.

N

N: Overview of sale of intermediates. Iodine-rich...

- ...biomatter can be sold to people as a vitamin supplement food or nutraceutical. This requires mass-marketing, a partnership with a vitamin manufacturer / spirulina company, a slew of safety checks, and advantage beyond current sources and ecology government credits.

With added incentive to us if buyers get sustainability credits:

- - ...biomatter can be added to animal feed as vitamin feedstock

- - ...biomatter can be added to fertilizer as organic course of biomass and minerals

- ...biochemical slurry can be sold to others who would further process it themselves into mineral iodine; the financial calculus is increased transport costs vs cost of further refining it in-house.

- ...biomatter can simply be dumped in landfill or compost if someone is paying for us to do iodine bioremediation of an overiodinated body of water

- ...biomatter can be sold for research purposes

- ...biomatter can be sold to crisis zones as an untasty but essential nutrient.

The quality, consistency, and iodine concentration (+low concentration of undesirables) of the biomatter or slurry are essential factors to solve for, as they set the standard for economic viability. Approval for consumption opens many markets.